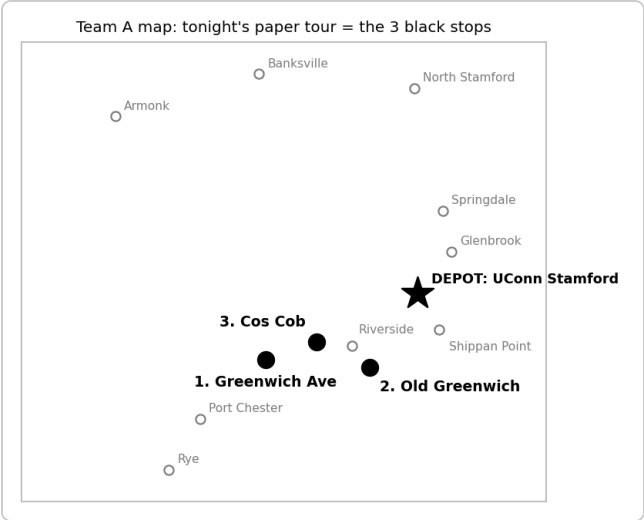


Studio 2 • The Stamford Delivery Tour

OPIM 5641 · Business Decision Modeling · work with your partner · pencil first, laptop second

Names: _____

Your van leaves **UConn Stamford**, visits **3 stops** once each, and comes back. Which order drives the fewest miles?



Driving miles (real roads, rounded)

miles	Depot	1. Greenwich Ave	2. Old Greenwich	3. Cos Cob
Depot	—	6	4	4
1. Greenwich Ave	6	—	5	2
2. Old Greenwich	4	5	—	4
3. Cos Cob	4	2	4	—

Read across: from the row stop to the column stop. Miles are the same in both directions.

1 • Every possible order

List all the orders (there are 6) using the **stop numbers**, e.g. *Depot → 2 → 1 → 3 → Depot*. Write the legs and add them up.

#	Order (Depot → ... → Depot)	Legs	Total miles
1			
2			
3			
4			
5			
6			

2 • What did you find?

Best tour: _____ = _____ miles. How many *different* lengths did you get? _____ Why? _____

3 • Predict before you code

3 stops = $3 \times 2 \times 1 = 6$ orders. How many orders for...

4 stops: _____ 5 stops: _____ 10 stops: _____ 12 stops: _____

If your laptop checks 250,000 orders per second, how long does 12 stops take? _____

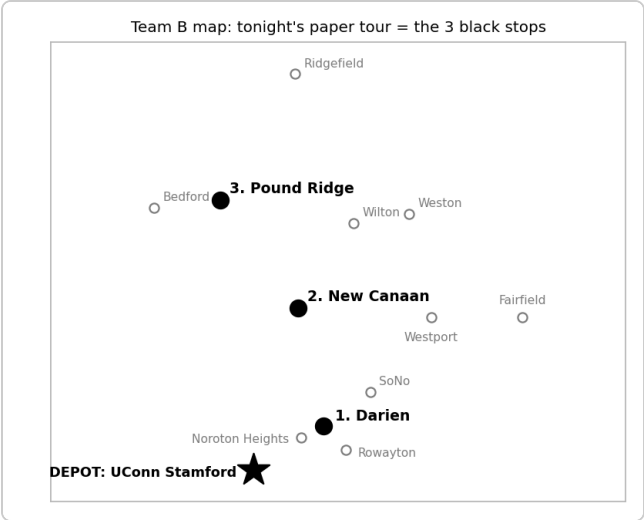
Next: open `Studio2_DeliveryTour_blank.ipynb`, set `TEAM = 'A'`, and check your table against the computer. Then scale up to 5 stops (the success) and all 12 (the wall).

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Names: _____

Your van leaves **UConn Stamford**, visits **3 stops** once each, and comes back. Which order drives the fewest miles?



Driving miles (real roads, rounded)

miles	Depot	1. Darien	2. New Canaan	3. Pound Ridge
Depot	—	5	10	12
1. Darien	5	—	5	13
2. New Canaan	10	5	—	7
3. Pound Ridge	12	13	7	—

Read across: from the row stop to the column stop. Miles are the same in both directions.

1 • Every possible order

List all the orders (there are 6) using the **stop numbers**, e.g. *Depot* → 2 → 1 → 3 → *Depot*. Write the legs and add them up.

#	Order (Depot → ... → Depot)	Legs	Total miles
1			
2			
3			
4			
5			
6			

2 • What did you find?

Best tour: _____ = _____ miles. How many *different* lengths did you get? _____ Why? _____

3 • Predict before you code

3 stops = $3 \times 2 \times 1 = 6$ orders. How many orders for...

4 stops: _____ 5 stops: _____ 10 stops: _____ 12 stops: _____

If your laptop checks 250,000 orders per second, how long does 12 stops take? _____

Next: open `Studio2_DeliveryTour_blank.ipynb`, set `TEAM = 'B'`, and check your table against the computer. Then scale up to 5 stops (the success) and all 12 (the wall).